



ANGELES UNIVERSITY FOUNDATION
Angeles City
COLLEGE OF COMPUTER STUDIES

COURSE SYLLABUS for PTF50

I. VISION	Inspired by the teachings and tradition of the Catholic Church, the Angeles University Foundation envisions to emerge as a Center of excellence in instruction, research, and community extension services in the region and in the global community.
II. MISSION	To realize its vision, the university is committed to the “Total Development of Man for God and Humanity”.
III. CORE VALUES	Mabuti (Inner Goodness). Initiate activities that will inspire the university-community to become authentic witnesses of the Christian Faith. Magaling (Excellence). Provide venues and opportunities for the university-community to become critical and analytical in issues that hinder the promotion of human dignity. May Malasakit sa Kapwa (Genuine Concern for Others). Prepare men and women who can translate their Christian principles into genuine compassion for others especially to the least, the last and the lost.
IV. INSTITUTIONAL LEARNING OUTCOMES	A. Values-Oriented (VO) B. Socially and Ethically Responsible (SER) C. Professionally Competent (PC) D. Critical and Creative Thinker (CCT) E. Lifelong Learner (LL) F. Globally-Oriented (GO)
V. COURSE DETAILS	Course Code: PTF50 Course Title: Professional Track 5: Web-Based AI Development (Data Science Track) Course Type: Lecture/Laboratory Credit Units: 3 Contact Hours: 5 hours per week Prerequisites: PTF04 Mode of Delivery: In Person Course Description: The course trains students to design, build, test, and deploy intelligent web applications that run machine learning models directly in the browser. Students will work with the modern web AI stack, namely TensorFlow.js, Transformers.js, ONNX Runtime Web, and MediaPipe Tasks, and will integrate generative AI services into web applications, with emphasis on practical data science workflows, privacy-preserving client-side inference, and responsible AI practices.

**VI. DESCRIPTION
OF THE TERMINAL
REQUIREMENT**

MCO1: Develop and deploy a web application that performs machine learning inference directly in the browser using a modern web AI framework.
MCO2: Design and develop a complete web-based AI application that integrates pretrained models and/or generative AI services to solve a real-world problem.

**VII. PROGRAM
LEARNING
OUTCOMES (PLOs)**

B.S. Computer Science

- A. Communicate effectively with the computing community regarding current trends in AI, data architectures, and machine learning, demonstrating a commitment to independent, life-long learning.
- B. Make effective presentations to a range of audiences about complex technical problems in data science and their solutions, utilizing clear technical documentation and data storytelling.
- C. Recognize the legal, professional, and ethical standards — including data privacy norms and public safety — in the utilization of computing technology, analyzing their impact on individuals, organizations, and society.
- D. Evaluate algorithmic principles and modern AI/computing tools (e.g., cloud platforms, deep learning frameworks) to select from a range of possible options, provide justification for that selection, and implement the solution in a specific data context.
- E. Apply computing fundamentals, mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of complex AI and data architectures.
- F. Evaluate, design, and develop scalable solutions — such as data pipelines and AI models — through software development and exploratory/experimental research for complex computing problems.
- G. Identify all the data, information, and knowledge elements, leveraging data mining and statistical modeling, to formulate solutions for computational science and machine learning applications.
- H. Work effectively and independently as an individual, and as a member or leader in diverse, multidisciplinary, and multi-cultural teams in software engineering and data operations.

**VIII. COURSE
LEARNING
OUTCOMES (CLOs)**

- 1. Explain the modern web AI landscape, including the architectures, capabilities, trade-offs, and ethical considerations of client-side and server-side AI deployment.
- 2. Build interactive web applications that perform machine learning inference in the browser using TensorFlow.js, Transformers.js, and ONNX Runtime Web.
- 3. Integrate pretrained vision and language models and generative AI services into web applications, applying data privacy and responsible AI standards.
- 4. Design, develop, and present a complete web-based AI application that solves a real-world computing problem.

IX. CURRICULAR MAPPING

MAPPING OF PLO and ILO

PLO	VO	SER	PC	CCT	LL	GO
PLO1						
PLO2						
PLO3						
PLO4						
PLO5						
PLO6						
PLO7						
PLO8						

MAPPING OF PLO and CLO

PLO	CLO1	CLO2	CLO3	CLO4
PLO1				
PLO2				
PLO3				
PLO4				
PLO5				
PLO6				
PLO7				
PLO8				

X. TOPICS AND TEACHING-LEARNING ACTIVITIES

LECTURE

Topic	Objective	Hours/Week	CLO	Teaching-learning activities		Assessment
I. The Modern Web AI Landscape A. Client-Side vs. Server-Side AI B. Browser ML Runtimes (WebGL, WebGPU, WASM) C. Privacy, Latency, and Offline Benefits D. Responsible and Ethical AI on the Web	- Describe the modern web AI landscape and its major frameworks. - Compare client-side and server-side AI deployment approaches. - Recognize ethical, privacy, and data protection standards in web AI applications.	6 hours 1st-3rd week	1	Teacher's activity	Class Discussion, Framework Demonstration	Short Quiz Knowledge Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
II. Machine Learning in the Browser with TensorFlow.js A. Tensors and Core Operations B. Pretrained Models (Image, Audio, Text) C. Transfer Learning in the Browser	- Understand the TensorFlow.js architecture and API. - Apply pretrained models in web applications. - Explain transfer learning using data collected in the browser.	6 hours 4th-6th week	1, 2	Teacher's activity	Class Discussion, Framework Demonstration	Short Quiz Knowledge Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
III. Natural Language Processing with Transformers.js	Understand transformer-based models and the pipeline abstraction.	4 hours 7th-8th week	2	Teacher's	Class Discussion,	Short Quiz Knowledge

A. Pipelines and Supported Tasks B. Hugging Face Models on the Web C. Sentiment Analysis, Summarization, and Zero-Shot Classification	• Select appropriate pretrained NLP models for web applications.			activity	Framework Demonstration	Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
Midterm Examinations						
IV. ONNX Runtime Web A. The ONNX Model Format B. Model Conversion and Optimization C. Execution Providers (WASM, WebGL, WebGPU)	• Understand the role of ONNX as an interoperable model format. • Explain how models are converted, optimized, and executed in the browser.	4 hours 10th-11th week	2	Teacher's activity	Class Discussion, Framework Demonstration	Short Quiz Knowledge Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
V. Real-Time Perception with MediaPipe Tasks A. Vision Tasks (Face, Hand, and Pose Landmarks) B. Text and Audio Tasks C. Designing Real-Time Interactive Applications	• Understand the MediaPipe Tasks API for vision, text, and audio. • Design responsive, real-time AI interactions for the web.	4 hours 12th-13th week	2, 3	Teacher's activity	Class Discussion, Framework Demonstration	Short Quiz Knowledge Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
VI. Generative AI Integration A. Large Language Model APIs and Prompt Design B. Safety, Cost, and Rate Limits C. Grounding Responses with Application Data	• Understand how generative AI services are integrated into web applications. • Apply responsible AI practices in prompt design and API usage.	6 hours 14th-16th week	3	Teacher's activity	Class Discussion, Framework Demonstration	Short Quiz Knowledge Check Assignment
				Students' activity	Seatwork, Recitations, Problem Solving	
VII. Final Project	• Presentation and defense of a complete web-based AI application.	2 hours 17th week	4	Teacher's activity	Project Consultation	Rubric Knowledge Check
				Students' activity	Project Presentation	
Final Examinations						

LABORATORY

Topic	Objective	Hours/Week	CLO	Teaching-learning activities		Assessment				
I. Web AI Development Environment A. VS Code, Node.js, and Live Server Setup B. JavaScript and DOM Refresher C. First In-Browser Inference Demo	- Set up a complete web AI development environment. - Run a first pretrained model directly in the browser.	9 hours 1st-3rd week	1	<table><tr><td>Teacher's activity</td><td>Hands-on Demo, Class Discussion</td></tr><tr><td>Students' activity</td><td>Hands-on Exercises</td></tr></table>		Teacher's activity	Hands-on Demo, Class Discussion	Students' activity	Hands-on Exercises	Hands-on Quiz Hands-on Exercises
Teacher's activity	Hands-on Demo, Class Discussion									
Students' activity	Hands-on Exercises									
II. TensorFlow.js Applications A. Image Classification App B. Webcam-Based Transfer Learning App	- Build web applications that classify images using pretrained models. - Train a custom classifier in the browser using webcam data.	9 hours 4th-6th week	1, 2	<table><tr><td>Teacher's activity</td><td>Hands-on Demo, Class Discussion</td></tr><tr><td>Students' activity</td><td>Hands-on Exercises</td></tr></table>		Teacher's activity	Hands-on Demo, Class Discussion	Students' activity	Hands-on Exercises	Hands-on Quiz Hands-on Exercises
Teacher's activity	Hands-on Demo, Class Discussion									
Students' activity	Hands-on Exercises									
III. Transformers.js Applications A. Sentiment Analysis App B. Text Summarization and Zero-Shot Classification App	Build NLP-powered web applications that run fully in the browser.	6 hours 7th-8th week	2	<table><tr><td>Teacher's activity</td><td>Hands-on Demo, Class Discussion</td></tr><tr><td>Students' activity</td><td>Hands-on Exercises</td></tr></table>		Teacher's activity	Hands-on Demo, Class Discussion	Students' activity	Hands-on Exercises	Hands-on Quiz Hands-on Exercises
Teacher's activity	Hands-on Demo, Class Discussion									
Students' activity	Hands-on Exercises									
Midterm Examinations										
IV. ONNX Runtime Web Applications A. Converting and Running a Model in ONNX Format B. Comparing Execution Providers	- Convert a trained model to ONNX and deploy it in a web application. - Compare inference performance across execution providers.	6 hours 10th-11th week	2	<table><tr><td>Teacher's activity</td><td>Hands-on Demo, Class Discussion</td></tr><tr><td>Students' activity</td><td>Hands-on Exercises</td></tr></table>		Teacher's activity	Hands-on Demo, Class Discussion	Students' activity	Hands-on Exercises	Hands-on Quiz Hands-on Exercises
Teacher's activity	Hands-on Demo, Class Discussion									
Students' activity	Hands-on Exercises									
V. MediaPipe Tasks Applications A. Hand and Face Landmark App	- Build real-time perception applications using webcam input. - Use landmarks and gestures to control application behavior.	6 hours 12th-13th week	2, 3	<table><tr><td>Teacher's activity</td><td>Hands-on Demo, Class Discussion</td></tr><tr><td>Students'</td><td>Hands-on Exercises</td></tr></table>		Teacher's activity	Hands-on Demo, Class Discussion	Students'	Hands-on Exercises	Hands-on Quiz Hands-on Exercises
Teacher's activity	Hands-on Demo, Class Discussion									
Students'	Hands-on Exercises									

B. Gesture-Controlled Web Application				activity		
VI. Generative AI Applications A. Chat Interface Using a Generative AI API B. Grounding Responses with Local Application Data	• Build a web application that integrates a generative AI service responsibly.	6 hours 14th-15th week	3	Teacher's activity	Hands-on Demo, Class Discussion	Hands-on Quiz Hands-on Exercises
				Students' activity	Hands-on Exercises	
VII. Final Project	• Implementation of the Final Project and Final Presentation.	9 hours 16th-18th week	4	Teacher's activity	Project Consultation	Hands-on Exam Rubric
				Students' activity	Hands-on Development, Presentation	
Final Examinations						

XI. GRADING SYSTEM

Midterm Period Raw = (Midterm Class Standing X 0.6) + (Midterm Exam X 0.4)
Midterm period raw is to be transmuted. The transmuted grade is encoded in School Automate as midterm grade.
Final Period Raw = (Final Class Standing X 0.5) + (Final Exam X 0.5)
Final Raw Grade = (Midterm Grade Raw + Final Period Raw) / 2
Final raw grade is to be transmuted. The transmuted grade is encoded in School Automate as final grade.

XII. REFERENCES (APA 6th or 7th edition style)

TensorFlow.js Guide and Tutorials (2026). Retrieved on July 9, 2026, from <https://www.tensorflow.org/js>
Transformers.js Documentation (2026). Retrieved on July 9, 2026, from <https://huggingface.co/docs/transformers.js>
ONNX Runtime Web Tutorials (2026). Retrieved on July 9, 2026, from <https://onnxruntime.ai/docs/tutorials/web/>
MediaPipe Solutions Guide (2026). Retrieved on July 9, 2026, from <https://developers.google.com/edge/mediapipe/solutions/guide>
Gemini API Documentation (2026). Retrieved on July 9, 2026, from <https://ai.google.dev/gemini-api/docs>
MDN Web Docs: JavaScript (2026). Retrieved on July 9, 2026, from <https://developer.mozilla.org/en-US/docs/Web/JavaScript>

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